

1. Evaluate the integral. $\int_1^{\infty} x e^{-x^2} dx$

Convergent to $\frac{1}{2e}$

2. Evaluate the integral. $\int_0^1 \frac{1}{(1-x)^2} dx$

Divergent

3. Does the sequence $a_n = \frac{8^{n+1} + 2^{n-2} + 5^n}{8^n}$ converge or diverge? If it converges, what does it converge to?

Convergent to 8

4. Consider the infinite series

$$21 - 15 + \frac{75}{7} - \frac{375}{49} + \frac{1875}{343} - \dots$$

Write this series in the form $\sum_{n=1}^{\infty} a_n$. Does this series converge? If so, find the sum.

$$\sum_{n=1}^{\infty} 21 \left(-\frac{15}{21} \right)^{n-1}$$

Convergent to $\frac{49}{4}$

5. Determine whether the series is convergent or divergent. Clearly state which convergence test you used.

$$\sum_{n=1}^{\infty} \frac{n^n}{(-4)^n}$$

Divergent

6. Determine whether the series is convergent or divergent. Clearly state which convergence test you used.

$$\sum_{n=2}^{\infty} \frac{3}{n(\ln(n))^2}$$

Convergent

-
7. Determine whether the series is convergent or divergent. Clearly state which convergence test you used.

$$\sum_{n=1}^{\infty} (-1)^n \frac{5n}{4n+7}$$

Divergent

-
8. Determine whether the series is convergent or divergent. Clearly state which convergence test you used.

$$\sum_{n=1}^{\infty} \frac{6n}{5n^2 + 4n + 7}$$

Divergent

-
9. Determine whether the series is absolutely convergent, conditionally convergent, or divergent. Clearly state which convergence test(s) you used.

$$\sum_{n=1}^{\infty} (-1)^n \frac{n^2}{2^n}$$

Absolutely Convergent

-
10. Determine whether the series is absolutely convergent, conditionally convergent, or divergent. Clearly state which convergence test(s) you used.

$$\sum_{n=2}^{\infty} (-1)^{n+1} \frac{n^2}{n^3 - 3}$$

Conditionally Convergent

11. For the following power series, find the radius and interval of convergence.

$$\sum_{n=0}^{\infty} \frac{3^n}{(n+1)^3} (2x-1)^n$$

$$\boxed{R = \frac{1}{6}, I = \left[\frac{2}{6}, \frac{2}{3}\right]}$$

12. Find the radius and interval of convergence.

$$\sum_{n=2}^{\infty} (-1)^n \frac{x^n}{4^n \ln n}$$

$$\boxed{R = 4, I = (-4, 4]}$$

13. Find the radius and interval of convergence.

$$\sum_{n=1}^{\infty} \frac{3^n (x+4)^n}{\sqrt{n}}$$

$$\boxed{R = \frac{1}{3}, I = \left[-\frac{13}{3}, -\frac{11}{3}\right)}$$

14. Find the power series representation of the function and determine the radius of convergence.

$$f(x) = \frac{x}{9+x^2}$$

$$\boxed{\sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{9^{n+1}}}$$

15. Find the power series representation of the function and determine the radius of convergence.

$$f(x) = \frac{x}{(1+4x)^2}$$

$$\boxed{\sum_{n=0}^{\infty} (-1)^n 4^n (n+1) x^{n+1}}$$

16. Determine the Maclaurin series for $f(x) = 4x^4 \cos(\pi x^5)$.

$$4 \sum_{n=0}^{\infty} (-1)^n \frac{\pi^{2n} x^{10n+4}}{(2n)!}$$

17. Find the **Taylor series** for $f(x) = \ln(x)$ centered at $a = 2$. Express your answer in summation notation.

$$\ln(2) + \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{2^n n} (x-2)^n$$

18. How many terms are needed to approximate the sum below to four decimal places?

$$\sum_{n=1}^{\infty} \frac{(-1)^n \cdot n}{8^n}$$

5 terms

19. How many terms are needed to approximate the sum below to ensure the error is less than 0.0000001 using the Integral Test Approximation?

$$\sum_{n=1}^{\infty} \frac{1}{n^2}$$

10,000,001 terms

20. Find the radius and ‘disk’ of convergence for the Complex Power Series below:

$$\sum_{n=1}^{\infty} 2^{2n} \left(z - \frac{1}{2}\right)^n$$

$$R = \frac{1}{4} \quad D = \left(\frac{1}{2}, \frac{1}{4}\right)$$